# Streamflow forecasting Models, datasets and completed work

LSTM • Stacked LSTM • Bidirectional LSTM • Transformer
Integrated using forecast Bayesian model averaging

| DATA USED | EXPERIMENT | VERIFICATION |
|---|---|---|
| 13,149 daily observations<br>USGS gauge 01646500 | 4 neural models + BMA<br>2,192 test forecasts | 10 tests passed<br>Saved predictions and audit files |

## Purpose and scope

The requested streamflow workflow has been implemented and run on a public USGS daily discharge series. This report explains the models, provides dataset source links, records the first experiment, and separates completed work from remaining research.

The original proposal concerns phosphorus retention in constructed wetlands. Its four cited data sources were investigated separately. With user approval, a fifth source was added for streamflow forecasting because the wetland sources did not provide an accessible, verified daily flow series suitable for this experiment.

## Main findings

**Stacked LSTM gave the lowest test RMSE: 111.22 m³/s.** BMA gave the lowest MAE: 37.79 m³/s, with RMSE of 111.87 m³/s. Equal averaging gave the highest KGE. The experiment therefore does not establish that BMA is best on every metric.

The BMA 90% prediction interval covered 92.97% of test observations. This is an initial, single-gauge, single-seed historical benchmark. It is not a deployed forecasting system or a completed validation of the wetland phosphorus proposal.

## Project record

GitHub project | Published implementation commit 83b3ae9

Implementation and experiment outputs were pushed to the main branch. This report summarizes that saved project state; numbers come from the recorded run rather than a new training experiment.

# Datasets cited in the proposal

Download status below refers to files actually obtained as of 8 October 2026. A source page, paper or metadata record is not counted as a downloaded dataset.

## 1. Webb et al. (2025) - downloaded

**Phosphorus Removal in Constructed Treatment Wetlands: A Systematic Review.** The supplementary ZIP and Table S2.xlsx were downloaded. The ZIP also contains Figures S1-S3 and Table S1 as PDFs.

Table S2 contains wetland design, monitoring, phosphorus and average-flow information. Its used range is 230 rows × 105 columns, including headers and a legend; this is not a count of 228 independent wetlands. The article synthesizes 71 studies. These are literature-derived wetland records, not the daily series used to train the forecasting models.

Article / DOI 10.3390/w17223301 | Supplement download page

Saved: data/raw/webb_2025/water-17-03301-s001.zip and Table S2.xlsx. Article license: CC BY 4.0.

## 2. Anderson et al. (2024) - not downloaded

**We know less about phosphorus retention in constructed wetlands than we think we do: A quantitative literature synthesis.** The paper identifies literature data and resampling-code supplements, plus an underlying Old Woman Creek dataset by McMurray et al. (2024), EDI package edi.1810.1.

Publisher and EDI access attempts encountered human-verification pages or HTTP 403. The data files were not obtained, and their variables, sampling structure and licensing remain unverified.

Anderson article / 10.1016/j.ecolind.2024.112969 | Old Woman Creek data / EDI DOI

## 3. UEA / Cooper (2020) - CSV identified; not downloaded

**Integrated Constructed Wetlands (ICW): Water Quality Data for the River Ingol and River Mun ICWs.** The catalogue describes monthly manual grab samples collected from 5 February to 19 September 2019 in Norfolk, UK. A 6.42 KB CSV is linked.

The CSV could not be saved because of access restrictions. Monthly water-quality samples are too sparse for the daily, 30-day-window experiment. Phosphate and total phosphorus must also be distinguished before any wetland data harmonization.

UEA dataset catalogue | Linked Wetland_Data.csv

# Remaining source and added flow data

## 4. Dykes (2025) - paper only; observations not obtained

**Impact of seasonal climate variability on constructed wetland treatment efficiency.** A university-hosted reference PDF was downloaded. Its data-availability statement says the underlying data are available on request. No observation dataset was downloaded and no author was contacted.

Article DOI | University-hosted article PDF

Local reference: data/raw/dykes_2025/article_reference_only.pdf. The reference PDF is excluded from Git. Its download must not be described as acquiring the study dataset.

## 5. USGS daily discharge - downloaded and used

**Station 01646500: Potomac River near Washington, DC, Little Falls Pump Station.** This public streamflow dataset was added with user approval and is the only dataset used for the reported neural-network and BMA experiment.

| PROPERTY | VERIFIED DATA |
|---|---|
| Period and frequency | 1 January 1990 to 31 December 2025; daily mean |
| Size and completeness | 13,149 daily observations; complete usable daily sequence |
| USGS selection | Parameter 00060: discharge; statistic 00003: daily mean |
| Units | Original $ft^3/s$ converted to $m^3/s$ using 0.028316846592 |
| Quality qualifiers | All observations approved (A); 174 also estimated (e), retained |
| Local files | Raw JSON preserved; processed streamflow.csv and audit JSON |

USGS station page | Direct daily-values JSON download | USGS data licensing

Retrieved 8 October 2026. Source: U.S. Geological Survey, National Water Information System. USGS-produced data are generally public domain, subject to source-specific notices. Checksums and source URLs are recorded in data/manifest.json.

## Suitability and boundaries

The long daily record supports chronological training, calibration and evaluation. The model uses historical flow and seasonality only. No rainfall, temperature, catchment attributes or wetland phosphorus measurements enter the experiment. The gauge is not assumed to represent unregulated or naturalized flow.

Historical observations may have been revised. This hindcast does not establish that the same approved values would have been available at each real-time forecast issue date.

# How the four models work

Each model observes the previous **30 days** and predicts the next daily mean discharge. Each day has three inputs: standardized log(1 + discharge), annual sine and annual cosine. A linear prediction head estimates a residual added to the latest standardized log-flow value.

## LSTM: gated memory through time

A Long Short-Term Memory network updates a cell state and hidden state at each day. Forget, input and output gates control which historical information is retained, written and exposed. This supports learning persistence, recession patterns and delayed changes in flow.

**Implemented:** one recurrent layer, 32 hidden units, final hidden state to the prediction head. **Tradeoff:** compact sequential model, but limited to information contained in the input window.

## Stacked LSTM: multiple recurrent layers

A second LSTM layer reads the sequence of hidden states from the first. The lower layer can encode short-term changes while the upper layer can combine them into more complex temporal features; such specialization is learned rather than guaranteed.

**Implemented:** two layers, 32 hidden units per layer, dropout 0.1 between layers; the upper final state feeds the prediction head. More depth adds capacity and can also increase overfitting risk.

## Bidirectional LSTM: two views of the observed window

One LSTM reads the historical window forward and another reads it backward. Their terminal states are concatenated, allowing the prediction head to use both summaries of the same observed history.

**Implemented:** one bidirectional layer, 32 units per direction. Both directions are restricted to the 30 past days, so the reverse pass does not expose the future target. Correct terminal forward and reverse states are used.

## Transformer: attention across past days

Self-attention compares days within the input window and learns which observations are relevant to one another. Multiple attention heads provide different learned comparisons; positional encoding preserves time order. Feedforward layers transform the attended representations.

**Implemented:** two encoder layers, width 32, four heads, feedforward width 64 and dropout 0.1. Sinusoidal position encoding is added; the last-token state feeds the head. All attention inputs are historical. Encoder-layer matrices are initialized independently.

## Training settings and reading

Shared settings: Adam, learning rate 0.001, batch size 128, up to 20 epochs, early stopping patience 5, gradient clipping 1.0, seed 42 and CPU execution. The loss is mean squared error in standardized log-flow space. Model parameter counts are not matched.

Hochreiter & Schmidhuber: LSTM (1997) | PyTorch LSTM reference | Vaswani et al.: Transformer (2017)

# Evaluation and Bayesian averaging

## Chronological partitions

| PARTITION | TARGET DATES | WINDOWS | PURPOSE |
|---|---|---|---|
| Training | 31 Jan 1990 - 31 Dec 2009 | 7,275 | Fit networks and scaling |
| Validation | 1 Jan 2010 - 31 Dec 2014 | 1,826 | Select network checkpoints |
| Calibration | 1 Jan 2015 - 31 Dec 2019 | 1,826 | Fit BMA and mean corrections |
| Test | 1 Jan 2020 - 31 Dec 2025 | 2,192 | Final recorded evaluation |

The first 30 days supply history rather than targets. Scaling is fitted using training data only. Incomplete or gappy windows are skipped, not interpolated. Forecasting is rolling one day ahead: observed past test flows may enter the next window, but the target and future flows never do.

## BMA working principle

The four models provide forecasts $f_k$ in unstandardized log(1 + flow) space. On the independent calibration period, each forecast receives a linear bias correction. BMA then combines four predictive distributions:

$$z = \log(1 + Q)$$
$$p(z \mid f) = \Sigma_k \, w_k \, \text{Normal}(z; a_k + b_k f_k, \sigma^2)$$
$$w_k \geq 0, \quad \Sigma_k \, w_k = 1$$

Correction coefficients a and b are fitted by least squares. Expectation-maximization (EM) estimates mixture weights and a shared residual variance by likelihood. Weights are not inverse-RMSE scores. The recorded fit converged after 1,815 iterations; shared log-space variance is 0.03408.

| LSTM | STACKED LSTM | BIDIRECTIONAL | TRANSFORMER |
|---|---|---|---|
| 0.027% | 76.706% | 6.695% | 16.572% |

Predicted flow is constrained to be nonnegative using Q = max(exp(z) - 1, 0). The implementation calculates the transformed mixture mean and 5th/95th percentile bounds. The point forecast is a discharge mean, rather than simply exponentiating the mean log forecast.

This is **empirical forecast BMA** with fitted network and calibration parameters. It does not integrate a full Bayesian posterior over neural-network weights. A small component weight describes calibration-period support; it does not prove an architecture is generally ineffective.

## Fair comparison baselines

Persistence predicts tomorrow equals today. Each individual model uses a calibration-period residual smearing correction when converting log forecasts to discharge means. Equal averaging takes the arithmetic mean of the four corrected point forecasts.

Raftery et al. (2005): forecast BMA / DOI 10.1175/MWR2906.1

# Recorded test results

USGS 01646500 | 1 January 2020 - 31 December 2025 | 2,192 daily next-day forecasts | Seed 42. Lower RMSE and MAE are better; higher NSE and KGE are better.

| MODEL | RMSE (m³/s) | MAE (m³/s) | NSE | KGE (2009) | BIAS (m³/s) |
|---|---|---|---|---|---|
| Persistence | 145.79 | 48.54 | 0.7619 | 0.8809 | +0.07 |
| LSTM | 121.31 | 40.71 | 0.8351 | 0.9128 | +6.75 |
| Stacked LSTM | 111.22 | 37.89 | 0.8614 | 0.9143 | +3.91 |
| BiLSTM | 121.57 | 40.29 | 0.8344 | 0.9143 | +5.75 |
| Transformer | 116.79 | 40.32 | 0.8472 | 0.9031 | +4.29 |
| Equal average | 115.14 | 39.44 | 0.8515 | 0.9197 | +5.17 |
| BMA | 111.87 | 37.79 | 0.8598 | 0.9160 | +3.34 |

**Interpretation:** stacked LSTM leads RMSE and NSE; BMA leads MAE; equal averaging leads KGE. BMA reduces RMSE by 23.26% relative to persistence, but its RMSE is slightly higher than stacked LSTM. No statistical significance or general superiority is established.

## BMA uncertainty and example hydrograph

The nominal 90% interval achieved **92.97% coverage**, with mean width **156.39 m³/s**. Coverage alone does not establish useful sharpness or reliable uncertainty at floods and low flows.

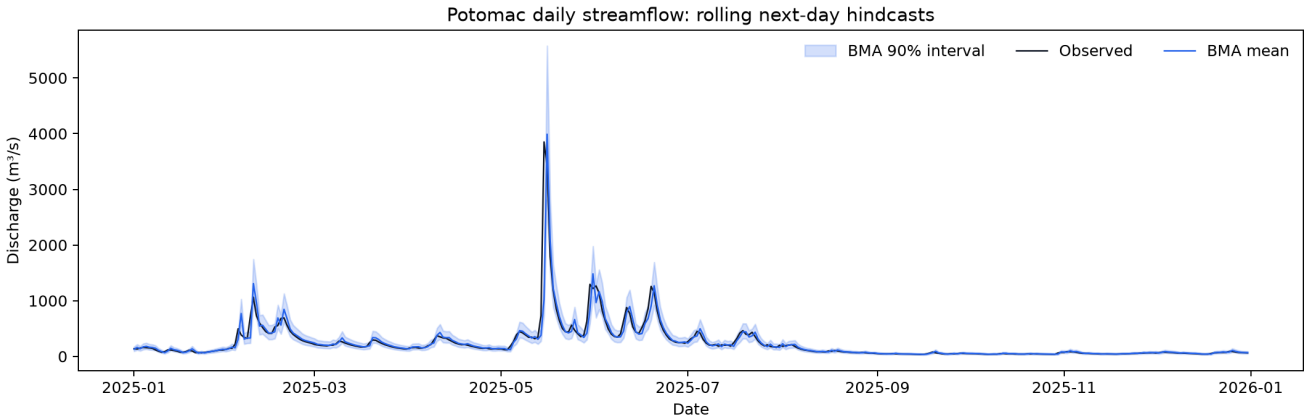



Figure 1. Saved 2025 hydrograph excerpt. The table above covers the full 2020-2025 test period.

## How to read the metrics

**RMSE:** square root of mean squared error; emphasizes large misses. **MAE:** mean absolute error. **NSE:** Nash-Sutcliffe efficiency; 1 is perfect and 0 equals the test-period observed-mean reference. **KGE (2009):** combines correlation, variability ratio and mean ratio; 1 is ideal. **Bias:** mean forecast minus observation; positive values indicate overprediction.

Evidence: results/baseline_seed42/metrics.json, test_predictions.csv, bma.json, run.json and test_hydrograph.png. Figures are rounded from saved values.

# Completed work and next steps

| WORKSTREAM | COMPLETED AND SAVED |
|---|---|
| Source audit | Four proposal sources investigated; Webb supplement obtained; Dykes paper obtained; access restrictions recorded. USGS daily series downloaded and audited. |
| Implementation | Data preparation, four neural architectures, forecast BMA, persistence and equal-average baselines. Local model checkpoints retained. |
| Experiment | Chronological training/validation/calibration/test run; daily prediction CSVs, metrics, training history, ensemble parameters and hydrograph saved. |
| Verification | 10 tests passed. Checks cover causal windows and scaling, gaps, gradients/batch independence, BMA weights and likelihood, interval ordering, transformed means, metrics and USGS quality filtering. |
| Publication | Source, documentation, dependency lock, raw USGS response, Webb files and experiment outputs pushed to GitHub main at 83b3ae9. Checkpoints and the Dykes reference PDF remain local. |

## Reproduce the saved workflow

From the repository root, create a Python 3.12 environment and use the pinned dependencies. The raw USGS response is included, so data preparation can run offline. Use a new output directory for a new experiment.

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
python -m hydrology.data
python -m pytest -q
python -m hydrology.train --output results/new_run
```

## What remains to be done

**Research validation:** multiple random seeds and gauges; rolling-origin evaluation; matched model capacities and a prespecified tuning budget. Several validation curves were still improving at the 20-epoch cap. Reserve a new untouched test set for future tuning.

**Forecasting and uncertainty:** add issue-time-available meteorological inputs and longer lead times; assess floods and low flows, proper probabilistic scores, interval reliability by flow regime, and sensitivity to excluding estimated observations.

**Wetland proposal:** obtain restricted/on-request data, verify licences and fields, harmonize phosphorus definitions and monitoring scales, and build the separate phosphorus-retention modelling workflow. Those proposal methods have not been implemented in this streamflow experiment.

The project currently establishes an executable historical benchmark and an auditable source inventory. It has not yet been prospectively tested or deployed.

Reproduction guide | Algorithm notes | Dataset audit | Saved experiment outputs